

8086 Assembly Language Programs

9. Addition of two 16-bit data

The screenshot shows the Online8086Compiler interface. The left pane displays the assembly code for `ADD16.ASM`, and the right pane shows the output console.

```
1 ; Program: Addition of two 16-bit numbers (8086)
2 DATA SEGMENT
3     NUM1 DW 1234H
4     NUM2 DW 4321H
5     RESULT DW ?
6 DATA ENDS
7
8 CODE SEGMENT
9     ASSUME CS:CODE, DS:DATA
10 START:
11     MOV AX, DATA
12     MOV DS, AX
13
14     MOV AX, NUM1
15     ADD AX, NUM2
16     MOV RESULT, AX
17
18     MOV AH, 4CH
19     INT 21H
20 CODE ENDS
21 END START
```

Output Console:

```
Assembling ADD16.ASM ...
0 Error(s), 0 Warning(s)
Running program in emulator ...
NUM1 = 1234H
NUM2 = 4321H
-----
AX = 5555H
RESULT = 5555H
Flags: CF=0 ZF=0 SF=0 OF=0
> Process exited normally.
```

10. Subtraction of two 16-bit data

The screenshot shows the Online8086Compiler interface. The left pane displays the assembly code for `SUB16.ASM`, and the right pane shows the output console.

```
1 ; Program: Subtraction of two 16-bit numbers (8086)
2 DATA SEGMENT
3     NUM1 DW 4321H
4     NUM2 DW 1234H
5     RESULT DW ?
6 DATA ENDS
7
8 CODE SEGMENT
9     ASSUME CS:CODE, DS:DATA
10 START:
11     MOV AX, DATA
12     MOV DS, AX
13
14     MOV AX, NUM1
15     SUB AX, NUM2
16     MOV RESULT, AX
17
18     MOV AH, 4CH
19     INT 21H
20 CODE ENDS
21 END START
```

Output Console:

```
Assembling SUB16.ASM ...
0 Error(s), 0 Warning(s)
Running program in emulator ...
NUM1 = 4321H
NUM2 = 1234H
-----
AX = 30EDH
RESULT = 30EDH
Flags: CF=0 ZF=0 SF=0 OF=0
> Process exited normally.
```

11. Multiplication of two 16-bit data

Online8086Compiler

Assembly (8086 / TASM)Run

MUL16.ASM

```
1 ; Program: Multiplication of two 16-bit numbers (8086)
2 ; 16x16 multiply gives a 32-bit result -> DX:AX
3 DATA SEGMENT
4     NUM1 DW 8000H
5     NUM2 DW 0002H
6     RESL DW ?      ; low word of result (AX)
7     RESH DW ?      ; high word of result (DX)
8 DATA ENDS
9
10 CODE SEGMENT
11     ASSUME CS:CODE, DS:DATA
12 START:
13     MOV AX, DATA
14     MOV DS, AX
15
16     MOV AX, NUM1
17     MUL NUM2      ; DX:AX = AX * NUM2
18     MOV RESL, AX
19     MOV RESH, DX
20
21     MOV AH, 4CH
22     INT 21H
23 CODE ENDS
24 END START
```

Output Console

Build Successful

```
Assembling MUL16.ASM ...
0 Error(s), 0 Warning(s)
Running program in emulator ...
NUM1 = 8000H
NUM2 = 0002H
-----
DX:AX = 0001:0000H
RESH (DX) = 0001H
RESL (AX) = 0000H
Flags: CF=1 OF=1 ZF=0 SF=0
> Process exited normally.
```

12. Division of two 16-bit data

Online8086Compiler

Assembly (8086 / TASM)Run

DIV16.ASM

```
1 ; Program: Division of two 16-bit numbers (8086)
2 ; DX:AX (dividend) / divisor -> AX = quotient, DX = remainder
3 DATA SEGMENT
4     NUM1 DW 0064H ; dividend = 0064H
5     NUM2 DW 0007H ; divisor = 0007H
6     QUOT DW ?
7     REM DW ?
8 DATA ENDS
9
10 CODE SEGMENT
11     ASSUME CS:CODE, DS:DATA
12 START:
13     MOV AX, DATA
14     MOV DS, AX
15
16     MOV DX, 0000H ; clear upper half of dividend
17     MOV AX, NUM1
18     DIV NUM2      ; AX = quotient, DX = remainder
19     MOV QUOT, AX
20     MOV REM, DX
21
22     MOV AH, 4CH
23     INT 21H
24 CODE ENDS
25 END START
```

Output Console

Build Successful

```
Assembling DIV16.ASM ...
0 Error(s), 0 Warning(s)
Running program in emulator ...
Dividend (DX:AX) = 0000:0064H
Divisor (NUM2) = 0007H
-----
AX (QUOT) = 000EH
DX (REM) = 0002H
Flags: ZF=0 SF=0
> Process exited normally.
```